Download and Extract HBase :

```
wget https://downloads.apache.org/hbase/2.5.11/hbase-2.5.11-bin.tar.gz
tar -xvzf hbase-2.5.11-bin.tar.gz
mv hbase-2.5.11 hbase
```

SETUP Environment variables:

```
echo 'export HBASE_HOME=/home/ubuntu/hbase' >> ~/.bashrc
echo 'export PATH=$HBASE_HOME/bin:$PATH' >> ~/.bashrc
source ~/.bashrc
```

XML CONFIGURATIONS:

```
<configuration>
  <property>
    <name>hbase.cluster.distributed</name>
    <value>true</value>
  </property>
  <property>
    <name>hbase.rootdir</name>
    <value>hdfs://localhost:9000/hbase</value>
  </property>
  <property>
    <name>hbase.zookeeper.property.dataDir</name>
    <value>/home/ubuntu/zookeeper</value>
  </property>
  <property>
    <name>hbase.zookeeper.quorum</name>
    <value>localhost</value>
  </property>
  <property>
    <name>hbase.unsafe.stream.capability.enforce</name>
    <value>false</value>
  </property>
  <property>
    <name>hbase.zookeeper.property.clientPort</name>
    <value>2181</value>
  </property>
  <property>
    <name>hbase.zookeeper.useExternal</name>
    <value>true</value>
  </property>
</configuration>
```

START HBASE:

```
ls -l /usr/local/hbase/bin/start-hbase.sh
```

start-hbase.sh

```
hbase:001:0> create 'student_info', 'info'
Created table student_info
Took 0.9485 seconds
=> Hbase::Table - student_info
hbase:002:0> put 'student_info', 'S1', 'info:name', 'Alice'
Took 0.1027 seconds
hbase:003:0> put 'student_info', 'S1', 'info:age', '22'
Took 0.0060 seconds
hbase:004:0> put 'student_info', 'S1', 'info:course', 'Computer Science'
Took 0.0033 seconds
hbase:005:0> put 'student_info', 'S1', 'info:grade', 'A'
Took 0.0031 seconds
hbase:006:0>
hbase:007:0> put 'student_info', 'S2', 'info:name', 'Bob'
Took 0.0040 seconds
hbase:008:0> put 'student_info', 'S2', 'info:age', '23'
Took 0.0026 seconds
hbase:009:0> put 'student_info', 'S2', 'info:course', 'Mathematics'
Took 0.0038 seconds
hbase:010:0> put 'student_info', 'S2', 'info:grade', 'B+'
Took 0.0070 seconds
hbase:011:0> get 'student_info', 'S1'
COLUMN                 CELL
 info:age              timestamp=2025-03-26T13:18:47.001, value=22
 info:course           timestamp=2025-03-26T13:18:47.022, value=Computer Science
 info:grade            timestamp=2025-03-26T13:18:47.044, value=A
 info:name             timestamp=2025-03-26T13:18:46.976, value=Alice
1 row(s)
Took 0.0239 seconds
hbase:012:0> get 'student_info', 'S1', 'info:name'
COLUMN                 CELL
 info:name             timestamp=2025-03-26T13:18:46.976, value=Alice
1 row(s)
Took 0.0207 seconds
hbase:013:0> scan 'student_info'
ROW                    COLUMN+CELL
 S1                    column=info:age, timestamp=2025-03-26T13:18:47.001, value=
```

```
1 row(s)
Took 0.0207 seconds
hbase:013:0> scan 'student_info'
ROW                    COLUMN+CELL
 S1                    column=info:age, timestamp=2025-03-26T13:18:47.001, value=
                       22
 S1                    column=info:course, timestamp=2025-03-26T13:18:47.022, val
                       ue=Computer Science
 S1                    column=info:grade, timestamp=2025-03-26T13:18:47.044, valu
                       e=A
 S1                    column=info:name, timestamp=2025-03-26T13:18:46.976, value
                       =Alice
 S2                    column=info:age, timestamp=2025-03-26T13:18:47.093, value=
                       23
 S2                    column=info:course, timestamp=2025-03-26T13:18:47.112, val
                       ue=Mathematics
 S2                    column=info:grade, timestamp=2025-03-26T13:19:00.198, valu
                       e=B+
 S2                    column=info:name, timestamp=2025-03-26T13:18:47.076, value
                       =Bob
2 row(s)
Took 0.0301 seconds
hbase:014:0> put 'student_info', 'S1', 'info:grade', 'A+'
Took 0.0133 seconds
hbase:015:0> delete 'student_info', 'S1', 'info:age'
Took 0.0114 seconds
hbase:016:0> deleteall 'student_info', 'S1'
Took 0.0037 seconds
hbase:017:0> disable 'student_info'
Took 0.6879 seconds
hbase:018:0> drop 'student_info'
Took 0.1437 seconds
hbase:019:0> brew services stop hbase
Traceback (most recent call last):
NameError (undefined local variable or method `hbase' for #<HBaseReceiver:0x20a2930f>)
Did you mean?  @hbase
hbase:020:0>
```